1. Project Name

Ukraine Food Security & Price Anomaly Assessment: AI-Based Early Warning for Food Price Spikes in Conflict Zones

2. Overview

This project aims to detect food security risks and price anomalies caused by the war in Ukraine using Machine Learning. By analyzing regional commodity price fluctuations together with conflict, currency, fuel and world-market data, the model provides an early warning mechanism for humanitarian aid organizations and decision-makers to intervene before food crises escalate. The project utilizes the World Food Programme (WFP) Ukraine Food Prices dataset, containing 5,136 cleaned records across 23 markets, product categories and regional administrative units (January–August 2026). These records were enriched with six additional sources:

- ACLED conflict events and UCDP geo-coded conflict data;
- the National Bank of Ukraine exchange rate;
- the IMF world wheat price;
- WFP diesel prices;
- the UN Black Sea Grain Initiative shipment records.

Exploratory Data Analysis (EDA) produced four key findings:

- The national food basket rose by 28.3% between January and August 2026, peaking in July, while diesel rose by 56.2% and the world wheat price by 40.5%.
- Price increases were almost uniform across oblasts (+23.5% to +33.0%), but very uneven across products: onions +226%, carrots +108%, and less than 3% for wheat flour, sugar and rice.
- Local conflict intensity was not followed by local price increases in Ukraine, whereas in Yemen it was followed by higher prices one month later.
- Domestic wheat flour and barley prices did not follow the rise in the world wheat price.

This project directly aligns with SDG 2 (Zero Hunger), specifically Target 2.1, which calls for ending hunger and ensuring access by all people to safe, nutritious and sufficient food all year round.

3. Background

Armed conflicts severely disrupt supply chains, local agriculture and market trade, leading to rapid price spikes and food insecurity. In conflict zones like Ukraine, real-time monitoring of essential food prices is critical to ensure timely aid distribution and prevent localized famines. Ukraine is also one of the world's largest grain exporters. Its Black Sea exports were blockaded in 2022 and partially reopened under the Black Sea Grain Initiative (August 2022 – July 2023), which was signed in Istanbul and under which every vessel was inspected in Turkish waters.

While traditional price monitoring relies on retrospective reporting, machine learning models can learn spatial, temporal and commodity-specific patterns, and can combine them with conflict and macroeconomic signals to highlight emerging market anomalies. This study

bridges this gap by measuring how conflict relates to food prices, and by building models that predict price levels and warn of price spikes one month ahead.

4. Key Objectives / Business Objectives

Research Questions

1. Conflict Lag: When conflict intensifies in a region, after how many months do local food prices rise?
2. Front-line Distance: How do markets near the front line differ from markets far from it?
3. Grain Corridor: What did the Black Sea Grain Corridor change, and what was Türkiye's role?
4. Regional Variance: How significantly do food commodity prices fluctuate across different regions (oblasts/markets) and products?
5. Model Efficacy: How accurately can a Random Forest model predict commodity prices (price_usd) and warn of price spikes one month ahead?
6. Transferability: Does the same approach work in other conflict zones (Syria, Yemen, Sudan)?

Key Steps

- Data Preparation: Load, inspect and clean the WFP Ukraine Food Prices dataset by handling missing values (reducing from 6,770 to 5,136 valid records).
- Data Integration: Merge conflict events (ACLED), distance to the front line (UCDP), exchange rate (NBU), world wheat price, diesel price and grain corridor shipments into a 184-row market × month panel.
- Feature Engineering: Extract temporal variables (year, month), encode categorical variables (market, commodity), and build a food basket index and monthly price changes.
- Analysis: Show price changes region by region, measure conflict–price lags, compare markets near and far from the front line, and analyse the grain corridor.
- Model Training: Train Random Forest models for price prediction and spike warning, and evaluate them with the original random split, a leave-one-month-out validation and a walk-forward validation.
- Transfer Test: Apply the same approach and the Ukraine-trained model to WFP data from Syria, Yemen and Sudan.
- Reporting: Integrate findings into a structured Concept Note for stakeholder reporting and capstone submission.

5. Methods and Workflow

a) Datasets

- WFP Ukraine Food Prices dataset: 5,136 cleaned records across 17 attributes (23 markets, 28 products, January–August 2026).
- ACLED Ukraine Conflict Events (HDX): monthly political violence events and fatalities by oblast (2025–2026).
- UCDP Georeferenced Event Dataset (HDX): 31,547 geo-coded conflict events, used to locate the front line.

- National Bank of Ukraine: official UAH/USD exchange rate, averaged by month.
- IMF Global Wheat Price (FRED): monthly world wheat price in USD/ton.
- UN Black Sea Grain Initiative (HDX): 1,004 voyages with port, cargo, tonnage, destination and Istanbul inspection date (August 2022 – July 2023).
- WFP Food Prices for Syria, Yemen and Sudan (same source and period) and ACLED data for each country, used for the transfer test.

b) Data Cleaning and Preprocessing

- Quality Assurance: Removed null values in key price columns (price_usd, local_price), yielding 5,136 clean observation rows.
- Temporal Extraction: Converted the date column to datetime objects, extracting year and month components.
- Categorical Transformation: Label encoded categorical fields including market (market_encoded) and commodity (commodity_encoded).
- Food Basket Index: Built from 19 food items. Each price was divided by its January 2026 value in the same market, and the items were combined with a geometric mean (January 2026 = 100).
- Data Integration: Matched WFP market names to ACLED oblast names. Created conflict lags of 0–6 months. Defined the front line as the UCDP grid cells with at least 50 events in 2023–2024, and computed each market's distance to the front in km.

c) Modelling

- Conflict Lag Analysis: A regression of monthly price change on local conflict events at lags of 0–6 months, with market and product × month fixed effects (3,052 observations).
    - No lag showed a significant positive effect; the only significant coefficient was negative, at 1 month (−0.74 pp, p = 0.007).
    - In the same analysis for other countries, Yemen showed a positive effect one month later (+2.96 pp, p < 0.01).
- Front-line Comparison: Markets within 100 km of the front were 3.0% above the national median price, against −2.5% for markets 100–300 km away and +1.6% for markets beyond 300 km. Their price changes were not more volatile: the standard deviation of monthly change was 15.0 pp, against 16.1 pp beyond 300 km.
- Grain Corridor: The corridor shipped 32.9 million tonnes in total, mainly corn (16.9 Mt) and wheat (8.9 Mt). Türkiye was the third-largest destination (3.24 Mt). Median inspection waits in Istanbul rose to 14–15 days in October 2022 and April 2023. In 2026, relative to the world wheat price (January = 100), domestic wheat flour and barley fell to 74–75, while millet and buckwheat, which are barely exported, rose to 107–112.
- Model Input Definition: Inputs X = [market_encoded, commodity_encoded, year, month, latitude, longitude] plus, in the new model, [conflict events (current and 1-month lag), national conflict change, distance to front, exchange rate, world wheat price, diesel price]. Target y = price_usd.
- Architecture: Random Forest Regressor configured with n_estimators=100, max_depth=15, random_state=42.

- Performance Metrics (price prediction):
    - Original model: MAE = $0.0423 USD and RMSE = $0.0840 USD with a random 80/20 split; MAE = $0.0616 USD and RMSE = $0.1206 USD when each month is predicted by a model that has not seen that month.
    - New model: MAE = $0.0622 USD and RMSE = $0.1169 USD with the same month-based evaluation.
    - Bread (rye): MAE = $0.0438 USD.
- Early Warning Classifier: Random Forest and Logistic Regression models predicting whether a product's price in a market will rise by at least 8.4% next month (the top 15% of monthly changes). Each month was predicted using only data from earlier months.
    - ROC-AUC = 0.62–0.64 and PR-AUC = 0.26–0.35, against 0.50 and 0.15 for a random guess.
    - At the selected alert threshold, the model catches 59% of spikes, and 30% of its alerts are correct (the random rate would be 18%).
    - The strongest signals were the national price change of the same product, its previous change and the exchange rate change.
- Transfer Test: A model trained only on Ukraine data achieved the following ROC-AUC values, compared with models trained on each country's own data:
    - Syria: 0.66 (own model: 0.66–0.67).
    - Sudan: 0.68–0.69 (own model: 0.62).
    - Yemen: 0.74–0.75 (own model: 0.72–0.74).

d) Deliverables

- Final Analyzed Dataset: The cleaned WFP Ukraine Food Prices data and a market × month CSV file combining food prices with conflict, distance to front, exchange rate, wheat price and diesel price.
- EDA & Visualizations: Statistical summaries, missing value patterns, region-by-region and product-by-product price trends, conflict lag effects, a near versus far comparison, and grain corridor and Türkiye analyses.
- ML Models: A Random Forest Regressor for price prediction and a Random Forest Classifier for food price spike warnings, together with a transfer test on Syria, Yemen and Sudan.

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